SECTION 3 · SUBSECTION 3.1
FILE INGESTION

3.1.1 INTRODUCTION.

Ingestion is the only stage that talks to GitHub. It obtains the complete path list for one branch, keeps the files that are worth reading, classifies the repository by size, and downloads the raw text of each kept file. It writes two large fields: a manifest of file metadata and a dictionary of path-to-text. The one idea the reader must leave with is that the manifest, not the repository, defines everything downstream; a file absent here cannot exist for later stages (figure 3.1-1).

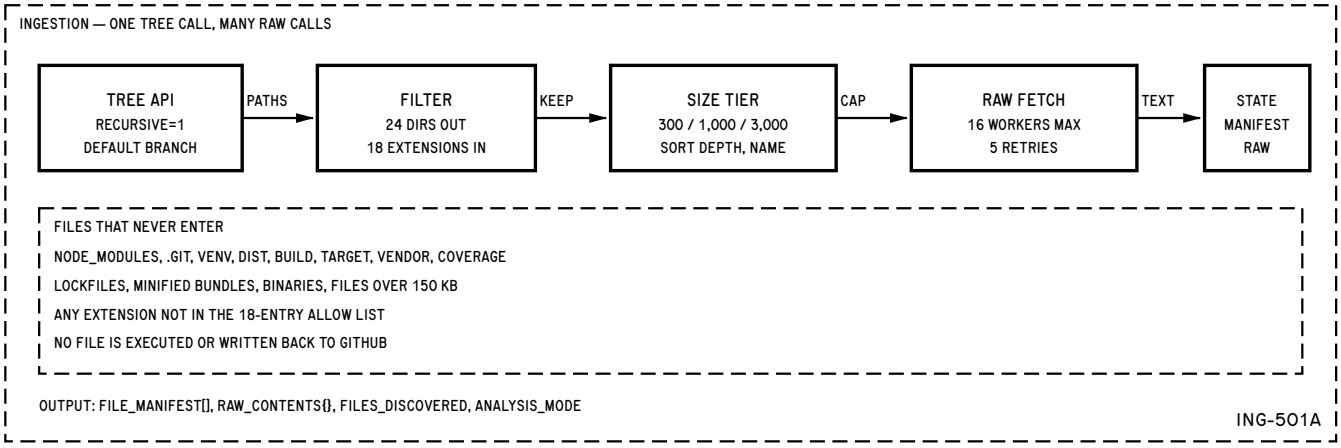


Figure 3.1-1. Ingestion Flow and Exclusion Boundary

3.1.2 FUNCTIONAL DESCRIPTION.

The stage calls *fetch_file_tree* once. For each entry it applies *should_include_file*, which checks directory segments first, then exact filenames, then minified suffixes, then the extension allow list, then the 150 KB ceiling. Surviving entries are sorted by slash count and then by path, so shallow files are preferred when a cap applies. The stage sets *analysis_mode* and calls *fetch_file_contents_batch*, which uses a shared requests session, a thread pool capped at sixteen, and five attempts with backoff. Line counts are computed from the returned text rather than trusted from the API.

3.1.2.1 Incremental Manifest Diff.

When *state.is_incremental* and a previous state exists, *compute_manifest_diff* classifies paths as added, modified, deleted or unchanged by SHA. Only changed paths are fetched; unchanged raw text is copied from the previous state when available. A deleted file is therefore visible to later stages as a diff entry, not as content.

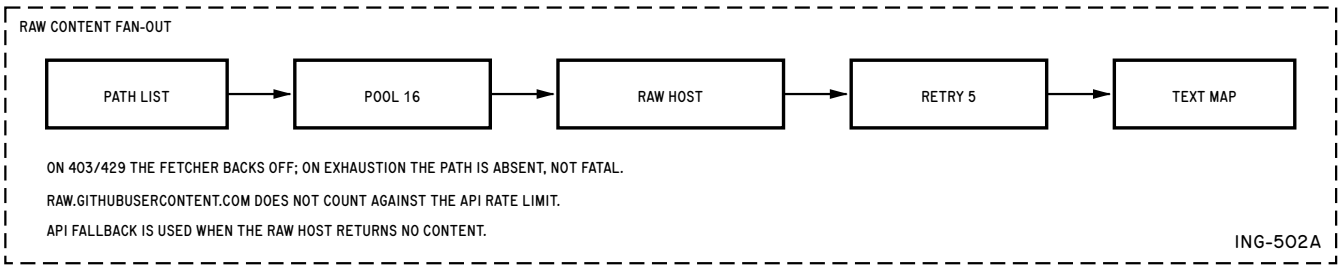


Figure 3.1-2. Content Fetch Pool and Retry Path

3.1.3 PERFORMANCE AND DESIGN DATA.

Table 3.1-1. Ingestion Parameters

Parameter	Value
Tree requests per run	1
Content workers	min(16, paths)
Fetch attempts	5, 0.5 s backoff
Inter-request delay	0.05 s
Maximum file size	150,000 bytes
Excluded directories	24 named patterns
Accepted extensions	18
Sort key	(depth, path)

CAUTION

The README says fifteen concurrent threads; the code caps at sixteen. It also lists a *tests/fixtures/sample_repo/* directory that is empty and untracked here. Trust table 3.1-1 and section 5.1-2, not the README tree.

INGESTION CHECK	OPERATOR
1 CONFIRM - GITHUB_TOKEN set	
2 READ - files discovered	
√ TIER - mode matches count	
3 CHECK - manifest not empty	
4 VERIFY - raw text for sample path	

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